

Can College Newspapers Save the Day?

A topic model analysis of critical information and community cohesion

Abstract: A limited news area (LNA) is a community with limited or no access to local news (Metzger 2025). This is detrimental since local news provides critical information as well as creates community cohesion (Khanom et al. 2023; Napoli et al. 2018; Waldman 2023). To combat this loss of local news, some colleges have started covering the local communities as opposed to just campus news (Bhowmik et. al. 2026). This research aims to discover what college papers discuss when covering the local community, specifically if it is critical information or community cohesion. We will also be looking at how a college's proximity to LNAs and news partnership status affects the topics of news articles. We will apply Latent Dirichlet Allocation (LDA) topic modeling to learn what college papers are discussing. We expect colleges to discuss community cohesion more than CIN. We also expect that colleges in or adjacent to LNAs are more likely to cover critical information needs (CIN) than those not in LNAs. Because of the difference in mission between news partnerships and regular student media, we expect that news partnerships will discuss CIN more regardless of their proximity to a LNA. College journalists cannot replace professions, so knowing what they are discussing and why can help provide quality news to millions of Americans.

Key Words: college news, community journalism, critical information needs (CIN), LDA topic modeling, local news, news deserts, news-academic partnerships, student media

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1) Introduction

Since the founding of the United States, the press has been seen as a public good that promotes freedom and democracy (Montgomery 2024, 7-8; Shaker 2014, 131, 133-134), yet there are around 50 million Americans without access to local news (Metzger 2025). Local news increases government transparency, voter turnout, and civil engagement by provided citizens with critical and community centered information (Abernathy 2020, 2, 11, 13; BU COM 2023; Cowan and Foster n.d., 3-4; Montgomery 2024, 18, 20; Napoli et al. 2018, 5; Pew Research Center 2016; Waldman 2023; Waldman 2026, 1-2; Watts 2023).

According to the Center for Community News (CCN), many student media organizations cover both campus news and the local community (Allen and Bhowmik 2025, 2). There are currently 766 college papers that are in or adjacent to an area with limited news, yet there has been no research on whether these colleges are more likely to cover community news that those in areas with robust news (CCN 2026; Lau 2026, 1). Student journalists are in a unique position to help produce local news, but they do not have the ability to completely replace professionals (Cowan and Foster n.d.; Watts 2023). It is important to understand what college papers are discussing when covering the community to learn what knowledge gaps still exist in limited news areas.

We aim to answer how student media discusses the community and what affect being located in or around an area with limited news has on the content of student media. First, we will review relevant literature regarding limited news areas, professional news, and college papers. Next, we will reiterate our research questions and expected results. Then we use Latent Dirichlet Allocation (LDA) to discover the topics in our corpus and create metatopics to simplify our analysis. We will interpret these results in the context of limited news areas and the possible role of college papers to understand how student media is helping communities with limited local news.

Finally, we conclude the paper by mentioning possible limitations with our analysis and areas for future research.

2) Literature Review

2.1) Limited News Areas

A news desert is an area or community with no access to local, reliable news. In addition to news deserts, there are also many counties with only one local news source (Metzger 2025). In this paper, we will be using the term limited news area (LNA) to refer to a community with zero or one local news source. As of 2025, there are 1,737 counties in the United States that are considered LNAs (Metzger 2025). Since the early 2000s, local newspapers have been financially struggling due to a decrease in print advertisements (Abernathy 2020) leading to 3,500 newspapers closing between 2005 and 2025 (Metzger 2025).

Research has also shown that many of the remaining papers have become shells of their former selves and are unable to provide communities with quality news (Abernathy 2020, 115; Metzger 2025; Watts 2023). Quality news is defined as original stories about the local community that meet citizens' critical information needs (CIN) (Napoli et al. 2018). While every community's CIN are different, Friedland et al. defined eight categories of information that are “necessary for citizens and community members to live safe and healthy lives” (Friedland et al. 2012, 6). Napoli et al. found that the average proportion of news stories that are original, local, and meet CIN is only 11.49% (Napoli et al. 2018).

In addition to covering CIN, journalists also create community cohesion by covering community events and people, like farmers' markets and obituaries (Cowan and Foster n.d., 3; McAdam 2025, 274; Montgomery 2024, 20; Waldman 2023). Cohesion journalism creates a sense of shared interest and identity within a community (Abernathy 2020, 13; Collier and Graham 2022,

3; Cowan and Foster n.d., 4; Khanom et al. 2023, 319; Montgomery 2024, 20; Shaker 2014, 133; Waldman 2023). Community cohesion is often overlooked when discussing journalism, but it is an important part of local news that is being lost (Cowan and Foster n.d., 3-4; Waldman 2023).

In the absence of traditional local news media, other sources — like citizen journalism, ethnic media, and community newsletters — have been providing quality news for their community (Collier and Graham 2022; Cowan and Foster n.d., 2; Montgomery 2024, 82; 2020; McAdam 2025; Stonbely 2025; Torre et al. 2024); this includes colleges and universities (Allen and Bhowmik 2025; Bhowmik et al. 2026; Blazina 2022; BU COM 2023; Cowan and Foster, n.d.; CCN 2025; Finn 2025; Finneman et al. 2022; Lau 2026; Marcus 2022; Setty and Galvão, n.d.; Watts 2023). The CCN found that almost 70% of student media organizations in their dataset regularly report on local news (Allen and Bhowmik 2025, 3). In addition to student media that is independently run by students, news-academic partnerships also provide communities with local news. News partnerships are “student reporting programs where faculty/staff review and edit student content before it is shared” (CCN 2025, 2). These partnerships cover the community and often work with existing media outlets to publish students' stories.

2.2) Topic Modeling

Many researchers have applied topic modeling to news articles to learn what is being discussed. Kretinin and Nguyen (2022) and Wang and Mengoni (2020) used Latent Dirichlet Allocation (LDA) to compared how topic proportions of news articles changed over time. Anna Verbytska (2024) used LDA and BERTopics to see how coverage of the Russia-Ukraine war differed between two news sites (Verbytska 2024). All three papers also focused on the parameter tuning process needed to find the best LDA topic model (Kretinin and Nguyen 2022; Verbytska 2024; Wang and Mengoni 2020). Ejaz et al. (2023) applied LDA to climate news articles extracting 50 topics (Ejaz

et al. 2023, 6-7). They further classified these into six meta topics manually, allowing for clearer analysis (Ejaz et al. 2023, 13-14).

Other researchers have analyzed news articles to see if they meet CIN. Napoli et al. used hand-coding to determine if news articles met a community's CIN as well as if the articles were local and original (Napoli et al. 2018). Khanom et al. used machine learning to assign news articles to CIN (Khanom et al. 2023, 323). The Local Journalist Index 2026 report used topic modeling to cluster news articles together then used Artificial Intelligence to assign each topic to a CIN (Muck Rack and Rebuild Local News 2026). They also employed the same method to assign topics to the International Press Telecommunications Council's (IPTC) media topic codes. IPTC media codes are a way to classify news articles by subject matter and are the standard taxonomy used by wire services (IPTC 2026; Muck Rack and Rebuild Local News 2026).

There have been many articles and reports about student journalists covering local issues, however, these studies are looking at a small number of schools (Allen and Bhowmik 2025; Bhowmik et al. 2026; Blazina 2022; BU COM 2023; Cowan and Foster, n.d.; CCN 2025; Finn 2025; Finneman et al. 2022; Lau 2026; Marcus 2022; Setty and Galvão, n.d.; Watts 2023). There is also a lack of research on the type of community news students are writing about (Allen and Bhowmik 2025, 6). We extended this research by conducting topic modeling on a large corpus of student written news articles. This helps fill the gap regarding the content of student articles and refines research on the number of student papers providing communities with critical news.

While many researchers have analyzed news articles using topic modeling, it has only been applied to professional publications (Ejaz et al. 2023; Meng 2025; Muck Rack and Rebuild Local News 2026; Verbytska 2024; Wang and Mengoni 2020). We help refine this research by applying

topic modeling to college papers. By combining these two areas of research, we hope to shed light on how student media may be decreasing the effects caused by LNAs.

On average, we expect colleges to discuss community cohesion more than CIN. Campus related stories, including sports, are considered community cohesion in our research, and we assume that is the primary topic most college papers write about. We expect that colleges in or adjacent to LNAs are more likely to cover CIN than those not in LNAs. It is also expected that colleges in communities with robust local news will cover community events that students may be interested in more than critical information. Because of the difference in mission between news partnerships and regular student media, we expect that news partnerships will discuss CIN more regardless of their proximity to a LNA. We will explore these hypotheses by answering the following research questions.

RQ1: Are college news articles more likely to cover CIN or build community cohesion?

RQ1.5: How does news partnership status and proximity to LNAs affect this?

RQ2: What do college papers write about in relation to their news partnership status and proximity to LNAs?

3) Methods

3.1) Data collection and cleaning

Our dataset contained 15,074 stories scraped from 904 student media organizations. To determine if these publications were in or adjacent to a LNA, we first found the college or university associated with each of the publications. To determine if it had a news-academic partnership, we used the CCN's list of news partnerships (UVM-CCN 2025). We then found the number of news sources in each college's county and the minimum number in the surrounding counties using the Schools & News Outlets 2023 dataset (CCN n.d.). If a university had multiple campuses and it

was unclear which one the publication belonged to, we assigned it the minimum number of news sources across campuses. For any college in our corpus that was not in Schools & News Outlets 2023, we used data from the Northwestern's Medill School of Journalism's 2024 report (Medill 2024). For publications that partnered with multiple institutions across a state, we assigned the adjacent number of news sources to the county with the lowest number of sources and left the number of sources for the publication's county blank.

A few of the publications were news partnerships whose scraped stories were not exclusively student written, so we dropped these publications. We then dropped any stories that were less than 150 words because they lacked the article or substantive content. We also dropped additional stories that did not contain actual content that we found during preliminary analysis. This left us with a corpus of 888 publications and 12,702 news stories. For further details on the data cleaning process and which publications were lost, see our available code and data.

In our corpus, 642 of the publications are located in or adjacent to a LNA producing 9,112 of our stories. In addition to that, 86 of the publications are a part of a news partnership. Table 1 further shows how proximity to LNAs is divided between the types of news partnerships for publications and stories.

Table 1: Publication and Story Count for News Partnership and LNA Status

News partnership	In a LNA	Adjacent to a LNA	Stories	Publications
No	No	No	215	2,922
		Yes	453	5,793
	Yes	Yes	124	1,617
		No	2	30
Yes	No	No	27	645
		Yes	53	1,483
	Yes	Yes	6	149

Table 1 shows the number of publications and stories for each combination of whether they are in a news partnership, located in a LNA, and adjacent to a LNA. Table 1 does not include any of the publications or stories with missing values. All non-listed combinations have zero publications and stories.

3.2) Latent Dirichlet Allocation (LDA) Preprocessing and Application

We applied LDA topic modeling which captures the hidden topic in a corpus of documents (Verbytska 2024, 175; Wang and Mengoni 2020, 764). LDA is based on the idea that each document has multiple topics in different proportions, and that each topic is represented by multiple words with different proportions that co-occur together in the documents (Ejaz et al. 2023, 6-7; Kretinin and Nguyen 2022, 250; Meng 2025, 335; Verbytska 2024, 175; Wang and Mengoni 2020, 764).

Before applying LDA, we preprocessed our data to ensure clean, interpretable topics (Wang and Mengoni 2020, 766). We applied standard preprocessing steps using the NLTK Python library but combined NLTK's stopwords with a more extensive list (University of Notre Dame n.d.). We also removed the name of the publications and institutions in our dataset as well as additional stopwords specific to our corpus. We used the Python library Gensim's predefined functions to create the LDA model. In the model, we removed words that were extremely frequent (in more than 90% of stories) or extremely rare (in less than 5 stories). Appendix A and our code contain further details about our preprocessing steps and model creation.

3.3) Finding the Best LDA Model

Before applying LDA topic modeling for analysis, we had to determine the best parameters for our dataset. We adjusted the number of topics (k), the topic density per document (α), the number of iterations, the number of passes, and the chunk size (Kretinin and Nguyen 2022, 250-251; Wang and Mengoni 2020, 766-767). We compared k and α together as done by Kretinin and Nguyen (2022). We then compared values for iterations, passes, and chunk size separately based on Wang and Mengoni's LDA method (2020, 766).

We used coherence scores to determine the best models. Coherence measures the similarity between words in the same topic (Meng 2025, 335) and has become a common measure of LDA topic modeling (Wang and Mengoni 2020, 764). We compared C_v , which was considered the best type of coherence by Röder, Both, and Hinneburg (2015, 405), with UMass to increase the validity of our findings (Wang and Mengoni 2020, 764).

We found that the best model for our corpus has 22 topics, an asymmetrical α , 50 iterations, 10 passes, and a chunk size of 500. Our chosen model has a C_v coherence score of 0.4568 and a UMass coherence score of -2.2771 . Further information regarding our tuning parameters, the coherence scores for each possible model, and how we decided on these values can be found in Appendix B.

3.4) Labeling the Topics

To label our topics we printed out the top 50 words for each of them as well as the two most representative articles. After reading the top words and articles, we assigned a specific label to each topic as well as an IPTC media code (IPTC 2026), metatopic, and whether it covered CIN or community cohesion. We also noted if any of the top words included words relating to colleges. We had 15 metatopics which consisted of the eight predefined CIN categories (Friedland et al. 2012; Muck Rack and Rebuild Local News 2026; Napoli et al. 2018) as well as six additional metatopics that represent community cohesion and a mixed category. The specific topic labels along with their IPTC, metatopic labels, and whether they are CIN or community cohesion can be seen in Appendix C. Appendix C also contains further details about the definition of each metatopic and how each topic label was decided.

We then aggregated the topics for clearer analysis. To do this we summed the topic proportions for all the topics categorized as CIN for each row, repeating for the community

cohesion topics and the mixed topics. This allowed each row in our aggregated topic dataframe to add to 1 (or 100%). We repeated this for the metatopics and IPTC topics resulting in three separate dataframes that all revealed what college papers discuss.

3.5) Statistical Analysis

We also performed statistical analysis on our results to strengthen our claims. To find if colleges discuss CIN and community cohesion at a different rate, we performed a non-parametric Wilcoxon signed-rank test on the difference between the amount community cohesion and CIN are discussed. We chose a non-parametric test because these differences were not normally distributed (Nahm 2016). We ran a one-sided test because campus related topics were categorized as community cohesion, and we expect college papers to discuss their campuses more than the community.

We then ran a Mann-Whitney U test to determine if news partnerships discuss CIN more than non-news partnerships. The proportion of CIN was extremely right skewed, so we once again ran a non-parametric test (Nahm 2016). This was also a one-sided test because we expected news partnerships would discuss CIN more than regular student media due to their mission. We ran the same test to see if LNA discuss CIN more than non-LNA since CIN proportions were still skewed. This was a two-sided Mann-Whitney U test as we did not have a substantial reason to believe LNA would discuss CIN more than non-LNA or vice versa. More information on the assumptions and tests can be found in our code.

4) Results

4.1) Are college news articles more likely to cover CIN or build community cohesion?

On average, the student written articles in our dataset discuss CIN 25.7% of the time. They discuss community cohesion, including campus related topics, 71.9% of the time, on average. The articles also discuss topics in our mixed category 1.8% of the time, on average. To confirm this, we ran a

one-sided Wilcoxon signed-rank test on the median difference between community cohesion and CIN. Our p-value of 0.0 provided substantial evidence in support of the claim that the median difference between community cohesion and CIN is greater than 0.0. This suggests that community cohesion is discussed more than CIN.

Figure 1: Average Proportion of CIN and Community Cohesion

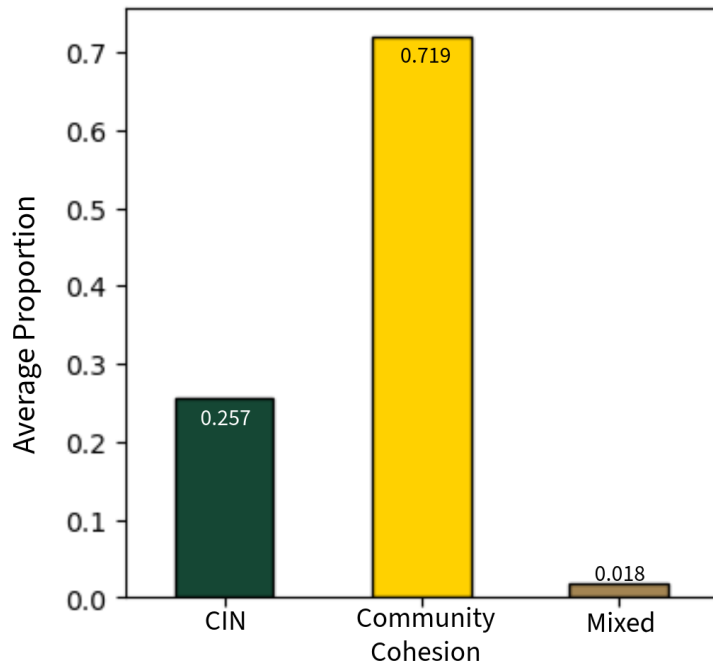


Figure 1 shows the average proportion CIN, community cohesion, and the mixed category are discussed in our data after aggregating the topics. Each bar is labeled with the aggregated topic group it belongs to (CIN, Community Cohesion, or Mixed) and with the first three decimals of its average proportion.

4.2) RQ1.5: How does news partnership status and proximity to LNAs affect RQ1?

Figure 2A shows that articles from publications not located in or adjacent to a LNA discuss CIN 27.5% of the time while those in or adjacent to a LNA discuss CIN 24.9% of the time, on average. For community cohesion topics, non-LNA articles discuss them 68.9% of the time, and LNAs discuss them 73.1% of the time, on average. We then ran a two-sided Mann-Whitney U test to determine if there was a statistically significant difference between the average rank sum amount LNAs discuss CIN and the average rank sum amount non-LNAs discuss CIN. With our p-value of 6.114e-06 we can reject the claim that the average rank sum of CIN discussed by articles from

LNAs and articles from non-LNAs discuss is the same. Based on our data, this suggests articles in or adjacent to a LNA discuss CIN less than articles not in or adjacent to a LNA.

Figure 2: Average Proportion of CIN and Community Cohesion Split by LNA and News Partnerships

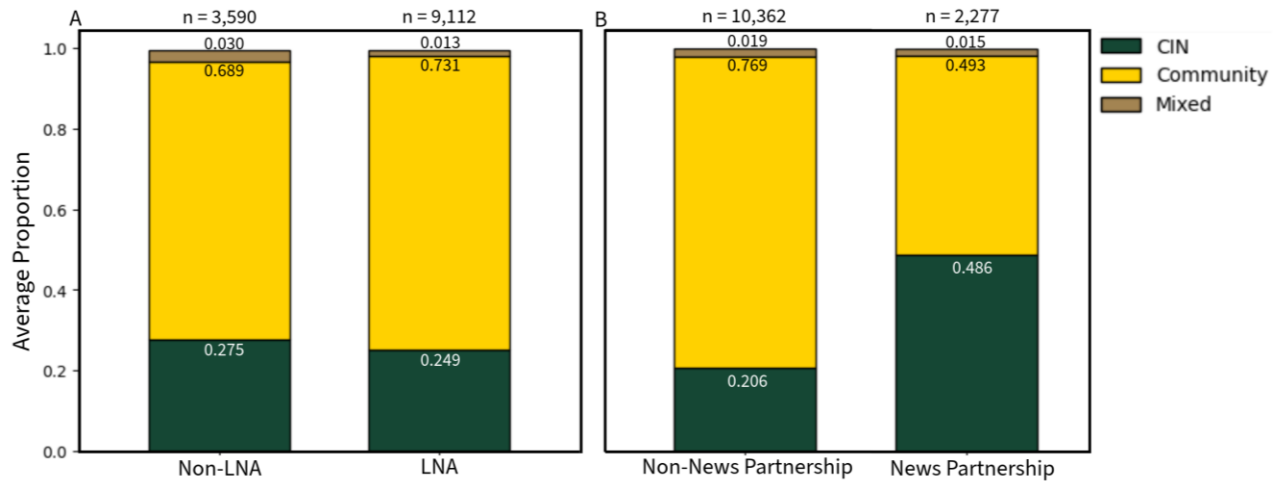


Figure 2A shows the average proportion CIN, community cohesion, and the mixed category are discussed for non-LNA and LNA. Figure 2B shows the average proportion CIN, community cohesion, and the mixed category are discussed for non-news partnerships and news partnerships. For both A and B, each bar is labeled with the group it belongs to, and each topic category (CIN, Community Cohesion, and Mixed in order from bottom to top) is labeled with the first three decimals of its average proportion for the specific group. The number of articles belonging to each bar is also labeled at the top of the figure as there is a large disparity between non-LNA and LNA as well as non-news partnerships and news partnerships.

Figure 2B shows the amount news partnerships and non-news partnerships discuss CIN, community cohesion, and the mixed category. Non-news partnerships discuss CIN 20.6% of the time and community cohesion 79.6% of the time, on average. Articles that are a part of a news partnership discuss CIN 48.6% of the time and community cohesion 49.3% of the time, on average. While the mean amount non-news partnerships and news partnerships discuss CIN is quite different, we still ran a one-sided Mann-Whitney U test to strengthen our results. Our p-value of $9.169e-259$ provided strong evidence in favor of the claim that the average rank sum non-news partnerships discuss CIN is less than the rank sum news partnerships discuss CIN.

4.3) What do college papers write about in relation to their news partnership status and proximity to LNAs?

Figure 3: The Average Proportion of Each Metatopic for News partnership and LNA Categories

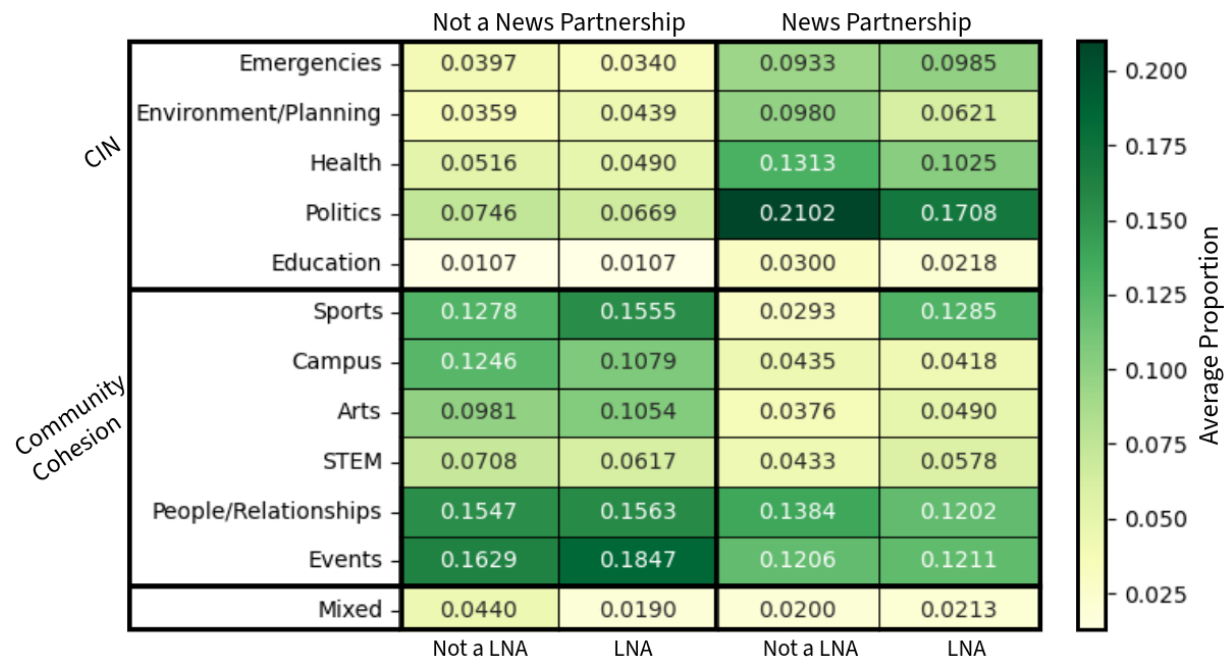


Figure 3 is a heatmap of the average topic proportions for each of the metatopics. The metatopics are divided into CIN and community cohesion (and mixed). A higher average proportion is represented by dark green while a lower proportion is light yellow. There are four columns of proportions which from left to right are non-news partnership and non-LNA, non-news partnership and LNA, news partnership and non-LNA, and news partnership and LNA.

Figure 3 shows that articles that are not a part of a news partnership discuss CIN at a low proportion, regardless of in they are in or adjacent to a LNA. The most commonly discussed topic for non-news partnerships is Events and People/relationships followed by Sports. The proportions for non-LNA and LNA that are not news partnerships are very similar, however, LNA do discuss Events and Sports more while non-LNA discuss Campus at a higher proportion. Proximity to a LNA seems to have almost no effect on what non-news partnerships discuss as the proportions for community cohesion topics are substantially higher than those for CIN.

For articles that are a part of a news partnership, they discuss both CIN and community cohesion at a similar proportion in total, but the average proportion per topic is higher for CIN. The most talked about CIN topic for news partnerships is Politics and the community cohesion topic with the highest proportions is People/Relationships and Events. There is more disparity between LNA and non-LNA within news partnerships than within non-news partnerships. For example, non-LNA discuss Environment/Planning more than LNA while LNA discuss Sports substantially more than non-LNA. Overall, Figure 3 shows that student media covers a large array of (meta)topics and that news partnership status has a larger effect on topic proportions than LNA proximity. We also created a similar figure with the IPTC media codes, however, we did not include it in the main paper as our metatopics were based on and align with the IPTC codes. This figure and our analysis of it can be found in Appendix D.

5) Discussion

We found that articles that are a part of a news partnership cover CIN more than those not in a news partnership, regardless of proximity to a LNA. This aligns with our expectations as news partnerships provide communities, not campuses, with news by definition. We also found that articles from publications located in or adjacent to a LNA discuss CIN less than those not in or adjacent to a LNA. This goes against our hypothesis that articles from LNA would discuss CIN more than articles from areas with robust local news. This suggests that colleges in LNAs are not stepping up and providing the community they are in with the critical information local newspapers are no longer able to provide. They may even be unable to produce as much CIN due to their proximity to LNAs, but more research is needed in this area. These results show that creating news partnerships may be the ideal strategy for increasing the amount of CIN covered by students and providing LNA with needed information.

Our results also revealed that college papers discuss both CIN and community cohesion but are more likely to cover topics relating to community cohesion. Our findings agree with our hypothesis that colleges would discuss community cohesion more because it includes campus, sports, and events students may be interested in. While news partnerships still discuss community cohesion topics more than CIN, the difference is less extreme than it is for non-news partnerships. This may be because news partnerships are discussing campus news at a lower rate than regular student media, or merely that they are producing community journalism more than CIN.

Overall, our results show that college papers are not primarily providing their communities with critical information, which is an essential need. However, building community and shared interest is also an important aspect of newspapers that colleges are helping to provide. Cohesion journalism has been seen as a role student journalists are especially fit for as the level of training needed is minimal (Cowan and Foster n.d. 3; Waldman 2023; Watts 2023). Student journalists are not able to replace professionals, but they are already positioned to become a community's main source of stories regarding community cohesion while professionals can focus more on CIN.

5.4) Limitations

The biggest limitation with our data is that the majority of it is from publications that are not a part of a news partnership and that are in or adjacent to a LNA. This means the averages across the whole dataset primarily represent this subgroup of student media. On the other hand, there are more publications that are not news partnerships than that are, and more publications in or adjacent to LNA than not (Bhowmik et al. 2026; CCN 2026; Lau 2026, 1). Therefore, our data may be more representative than an even sample of each of the four possibilities. This does not discredit our study or change our results but is something to consider.

The biggest limitation with our analysis is that we did not account for if articles were about the community, whether CIN or cohesion, or about the campus. There was a Campus metatopic, but this only captured words directly relating to campuses and universities/colleges. Other metatopics, like Sports, People, Arts, Events, and even some CIN metatopics could have been about a campus not the local community. We would have ideally coded each article as campus, local, state, national, or international and then only conducted our analysis on the local ones. This would have allowed for a more in-depth analysis of what college papers are discussing when discussing the local community.

5.5) Future research

One of the biggest areas where this research could be extended is by only looking at articles that are directly discussing the community. As mentioned above, we would have ideally labeled the locality of each article and then performed topic modeling and our analysis on only local news stories. This would also allow for analysis on the proportion of student written articles that are about campus, the local community, or a broader community in relation to their news partnership status and proximity to a LNA.

Another area for possible research could be to apply a different type of topic modeling on the data. While we chose LDA for its balance of quality and speed, there are other possibilities. BERTopic is a fairly common topic model with high quality, but is computationally complex (Meng 2025, 337). We also chose LDA because it assumes each document is composed of multiple topics; however, other studies have used BERTopics to assign only one topic to each article (Verbytska 2024). It would be interesting to compare the results of two different models and see how they may differ.

6) Conclusion

In a time when newspapers are closing rapidly and millions of Americans do not have access to reliable news, it is important to understand how college papers can help. Our study helps address the gap in the literature regarding what student papers write about. We found that college papers discuss topics relating to community cohesion more than those addressing critical information.

College papers do cover some CIN, especially if they are a part of a news partnership, while being in or adjacent to a LNA actually decreases the amount of CIN discussed. Even though we did not exclusively analyze articles that discussed community news, this research help shed light on what news college papers are providing their communities with. Overall, we discovered that college papers are more suited to providing local communities with cohesion journalism.

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Appendix A: *Preprocessing the Data for LDA*

During preliminary topic modeling, we noticed many of the topics contained written numbers, most likely due to scores and rankings in sports. We decided to drop written numbers from one to nine hundred and ninety-nine since they did not have meaning out of context. We had already dropped digits from 1-999 (Wang and Mengoni 2020, 766) and did not see a reason to keep the written version. We kept ordinal numbers (first, second, etc.) as these do contain contextless meaning.

We also dropped the words 'title', 'url', 'source', 'published', 'extracted', and all links. Each of the text documents contained these words and a link in their header meaning they would be overrepresented in the corpora and have context unrelated to the news articles. While these words would have most likely been dropped while creating the LDA model due to their frequency, we decided to manually remove them as well. We also dropped links since they did not contain context, and every document had a distinct one. We also noticed words from these links (like com and org) appearing in topics during our preliminary model analysis.

During our preliminary topic modeling, which used a smaller data set, we noticed many of the top documents were from the same publication, even if they were not about similar news stories. We also saw that most of the top documents were coming from the same publications, suggesting they may be overrepresented in our model. This was most likely caused by publications, institutions, locations, or the uneven number of stories from each publication shaping topics instead of the content. We reran our model with 25 and 50 topics after dropping publications and institutions. For 25 topics, we found that there were more publications in the most representative documents and less topics with documents from the same publication. We also noticed that when the most representative documents were from the same publication, they were often discussing the

same or very similar events. We found similar results when looking at 50 topics, however, there were more repeating publications, and some topics seemed to be grouping based on location only.

We then dropped the names of all U.S. states and Washington D.C. as well as the city each publication/institution is in. For both 25 and 50 topics, there seemed to be more repeating publications. When both the top documents were from the same publication, they either were discussing the exact same thing or did not seem related at all. The models without locations also had better UMass coherence, but worse C_v coherence. Because of this, we decided to only drop the publication and institution names during preprocessing. It provided a more diverse set of publications for top documents and had less topics with both top documents being from the same publication without losing topic coherence.

Appendix B: Tuning the LDA Model

Parameter tuning and coherence scores were used to find the ideal LDA model for our corpus.

K and α :

To find the best k, we first compared values from 10 to 100, increasing by 10 for both symmetrical and asymmetrical α . When α is symmetrical, every topic has the same distribution of $\frac{1}{k}$ while asymmetrical means each topic has a distribution of $\frac{1}{t+\sqrt{k}}$ where t is the index of the individual topic. We ran a LDA model with 5 passes, 10 iterations, the default chunk size of 2,000, and the changing α and k. The C_v and UMass coherence did not agree for many of these models with the top two UMass coherence scores being the bottom two for C_v . The third highest UMass score was asymmetrical alpha with 20 topics which was the eighth highest C_v score. That is the first time there was any overlap between the top models for the two coherence scores. Because of this we decided the model with 20 topics and an asymmetrical α was the best for our corpus as seen in Table 2

Table 2: Coherence Scores for Different Alpha and Number of Topics

α	K	C_v	UMass	α	K	C_v	UMass
Symmetrical	10	0.354	-1.039	Asymmetrical	10	0.353	-1.038
	20	0.371	-1.122		20	0.374	-1.115
	30	0.369	-1.206		30	0.373	-1.184
	40	0.368	-1.152		40	0.368	-1.150
	50	0.371	-1.140		50	0.373	-1.155
	60	0.375	-1.202		60	0.377	-1.218
	70	0.375	-1.278		70	0.379	-1.294
	80	0.374	-1.297		80	0.376	-1.305
	90	0.382	-1.316		90	0.381	-1.365
	100	0.372	-1.461		100	0.368	-1.500

Table 2 shows the C_v and UMass coherence scores, rounded to the nearest thousandth, for every value of k from 10 to 100 increasing by 10. The left side of Table 2 shows these values for a symmetrical α , and the right side shows them for an asymmetrical α .

We then looked at asymmetrical topic numbers from 15 to 25 increasing by 1. This allowed us to further identify the best k value based on our preliminary findings (Wang and Mengoni 2020, 766). As seen in Table 3, 22 topics had the highest UMass and the second highest C_v score. The number of topics with the highest C_v is 20, which is the fourth highest for UMass. We decided that 22 topics with an asymmetrical α was best for our corpus.

Table 3: Coherence Scores for Different Number of Topics

K	C_v	UMass	K	C_v	UMass
15	0.328	-1.129	21	0.330	-1.111
16	0.329	-1.122	22	0.330	-1.091
17	0.330	-1.096	23	0.328	-1.119
18	0.327	-1.120	24	0.329	-1.144
19	0.332	-1.116	25	0.329	-1.132
20	0.331	-1.115			

Table 3 shows the C_v and UMass coherence scores, rounded to the nearest thousandth, for k values from 15 to 25 increasing by 1 with an asymmetrical α . The chosen k is highlighted.

Iterations, Passes, and Chunk Size:

We looked at values of 10, 50, 100, and 500 for iterations and chunk size, and 5, 10, 50, and 100 for passes (Wang and Mengoni 2020, 767). Iteration controls the maximum number of times to iterate through a document when creating topic proportions, passes is how many times to train on the whole corpus, and the chunk size is the number of documents in each training chunk (Wang and Mengoni 2020, 764).

For iterations, we ran a model with 22 topics, an asymmetrical α , 5 passes, and a chunk size of 2000. The highest UMass was 10, but it had the lowest C_v . The highest C_v was 50 iterations, which was the second best UMass, as seen in Table 4. Because of this, we decided 50 iterations was best for our corpus.

We ran a model with 22 topics, an asymmetrical α , 50 iterations, and a chunk size of 2000 to find the ideal number of passes. As seen in Table 4, the best C_v is the worse UMass and vice

versa. There is also no agreement on the second best number of passes. The difference between 10 and 50 passes for UMass is about 0.10 while the difference for C_v is about 0.08. Because the difference is large for UMass, we went with the second highest coherence UMass score as the best model which was 10 passes.

For chunk size, we ran a model with the other predetermined ideal parameters, only adjusting the chunk size. The highest UMass coherence was 100 chunks, but that was the lowest C_v score, as seen in Table 4. The highest C_v score was 500 chunks, which had the third highest UMass score. Because of this, we decided 500 chunks balanced the two coherence scores the best.

Table 4: Coherence Scores for Different Values of Iterations, Passes, and Chunk Size

Iterations			Passes			Chunk Size		
Parameter values	C_v	UMass	Parameter values	C_v	UMass	Parameter values	C_v	UMass
10	0.372	-1.092	5	0.407	-1.259	10	0.407	-1.549
50	0.407	-1.259	10	0.441	-1.627	50	0.386	-1.319
100	0.405	-1.339	50	0.521	-1.730	100	0.378	-1.270
500	0.401	-1.350	100	0.541	-1.731	500	0.412	-1.394

Table 4 shows the C_v and UMass coherence scores, rounded to the nearest thousandth, for each of the tested parameter values for iterations, passes, and chunk size. The chosen parameter value is highlighted for each of the three parameters.

Appendix C: *Labeling the topics*

While there are seventeen IPTC media codes, we only had topics that aligned with nine of them. The seventeen codes, a description of them, and their subcategories can be found on the IPTC's website (IPTC 2026). For our project we assigned any topic relating to college campuses to the education code, which is not completely aligned with its definition but works for the purpose of our study.

We used Muck Rack and Rebuild Local News' definition of the eight CIN categories to determine if our topic aligned (2026). A short description of our six community metatopics can be seen below. Any topic that did not fit into the predefined CIN categories was considered community cohesion.

The Metatopics:

Art: Includes visual and performance art, like music, movies, painting, fashion, and more. This closely aligns with the Arts and Entertainment subtopic under the IPTC Arts, Culture, Entertainment, and Media topic.

Sports: Includes any form of competitive or non-competitive sports. This closely aligns with the IPTC Sport topic.

Science, Technology, Engineering, and Math (STEM): This topic contains any discussion of science and technology that is not a CIN. Includes different types of STEM subfields as well as computers, the internet, and social media. This closely aligns with the Science and Technology IPTC topic, however, it also includes social media which is a part of the Arts, Culture, Entertainment, and Media topic.

People/Relationships: Includes individual people, their accomplishments, and interpersonal relationships between people.

Community Events: Includes public events held for a community or its subgroups. These events are about bringing people together and/or celebrating, not about informing the community about a CIN.

Campus: Includes any topics that are exclusively about colleges, their campuses, and their students.

We also recorded if some of the top 50 words were directly related to colleges and campuses. This included words like college, university, dorms, major, etc. While this does not mean these topics were given a label related to colleges, it does allow us to learn more about which topics were closely linked to colleges.

Labeling the Topics:

Each of the topics, each of the labels given, and a short description of why can be seen below. The top five words and a link to the top two articles is also below. The full list of top 50 words and their proportions within the topic can be found in our code.

Topic 1: Pregnancy, weather, media

Top 5 words: paul, yeah, abortion, snow, kassandra

Top 2 stories:

1. <https://www.talonmarks.com/multimedia/2026/05/19/certified-chismosas-podcast-the-d4vd-case/>
2. <https://iowastatedaily.com/339760/news/iowa-state-dairy-farm-open-house/>

IPTC: None

Metatopic: Other

CIN or Community Cohesion: Mixed

Campus words: No

The top words for this topic are all over the place. Some are about pregnancy/abortion, weather/days, location, and characters/names. The top document is a podcast with a transcript. The podcast is about a murder and mentions pregnancy. It is also hosted by Paul, Cassandra, and Abby which are all top words. Story 2 is about an open house at an Iowa dairy farm. There are very few connections between the top words and/or the top stories. We will be categorizing this as none/other since a clear label cannot be decided on.

Topic 2: Minority communities and nature

Top 5 words: people, community, city, history, queer

Top 2 stories:

- 1) <https://vtcommunitynews.org/2026/06/26/foresters-adapt-as-insects-become-driving-force-in-tree-mortality/>
- 2) <https://www.theprospectordaily.com/2026/06/19/juneteenth-el-paso-mayor-renard-johnson-chris-canales-black-el-paso-voice/>

IPTC: Society

Metatopic: People/Relationship

CIN or Community Cohesion: Community Cohesion

Campus words: No

The top words are about communities and events regarding different minorities as well as nature. There are words that connect to LGBTQIA+, Native Americans and Black Americans. Story 1 is about how insects are causing many trees in VT to die. Story 2 is about a Juneteenth celebration in El Paso, Texas. It mentions the significance of the day and details about the celebration. These two stories are not related, and the top words cover both nature and minority communities. The biggest connection is either that Story 2 talks about foresters as a group, or nature needs rights and protection similar to people. Since most of the top words are not about nature, but communities that is our main label. For the metatopic, we were between people/relationships and events, because the second story is not about a specific event, but trees/foresters; we went with people as the metatopic.

Topic 3: Sustainability

Top 5 words: people, community, fashion, year , time

Top 2 stories:

1. <https://regis-highlander.squarespace.com/home/2026/4/28/the-misconception-behind-the-environmental-impact-of-thrifting?rq=The%20Misconception%20Behind%20the%20Environmenta>
2. <https://www.bradleyscout.com/news/viral-is-taking-on-a-new-meaning-in-chicken-coops-and-the-bradley-campus/>

IPTC: Environment

Metatopic: Health and Welfare

CIN or Community Cohesion: CIN

Campus words: Yes

The top words are about sustainability/the environment, campus, and health. These do not relate, especially health, since campuses can be connected with sustainability. Story 1 is about thrifting and how overconsumption can reduce its sustainability. Story 2 is about the Bird flu and its possible harms to the poultry industry. Both stories are about the environment which agrees with many of the top words. We will label this topic sustainability even though it disregards some of the top words. Since the words relating to the environment are not about natural disasters, the *Emergencies and Risks* CIN metatopic does not seem like the best fit. We were between Civic

Information and Health and Welfare as the metatopics. We decided on Health and Welfare because some of the top words perfectly fall into this category and being sustainable is necessary for long term welfare.

Topic 4: Sports (NFL Draft)

Top 5 words: draft, pick, msu, nfl, selection

Top 2 stories:

1. <https://merrimacknewspaper.com/countdown-to-chaos-nfl-draft-is-just-around-the-corner/>
2. <https://merrimacknewspaper.com/draft-night-dreams-predicting-the-2026-nba-draft/>

IPTC: Sport

Metatopic: Sports

CIN or Community Cohesion: Community Cohesion

Campus words: Yes

The top words are about the NFL draft, other sports, universities/cities, and mascots. Story 1 is about the 2026 Draft and some big questions and predictions. Story 2 is also about the NFL Draft and the author's theory on who will be drafted first/ their ideal draft. Because the top words also discussed other sports (NBA, sports), we are not just labeling this NFL or NFL draft. Because it is a mix of sports and level, we are just calling it sports.

Topic 5: Housing (Spanish)

Top 5 words: campus, building, housing, los, students

Top 2 stories:

1. <https://caplinnews.fiu.edu/crisis-vivienda-estudiantil-miami-ojeda-fiu-apartamentos/>
2. <https://nmsuroundup.com/33293/parriba/proprietarios-de-tierras-en-texas-temen-que-el-gobierno-federal-las-modifique-o-se-las-expropie/>

IPTC: Economy, Business, and Finance.

Metatopic: Environment and Planning

CIN or Community Cohesion: CIN

Campus words: Yes

The top words are about student housing and natural disasters, plus some Spanish words. Both of the stories are in Spanish, but we translated the title for some information. Story 1 is about how Miami is one of the most expensive U.S. cities for college students. Story 2 is about land in Texas and the federal government altering private land. This topic has a few different subset, but it is mostly about housing (and Spanish articles). This topic does not have a clear metatopic or IPTC topic. For the metatopic, we were between Emergencies and Risk because of the natural disaster words and Environment and Planning because of the housing words. Because Environment and Planning includes housing, land use, and environmental issues, we went with this metatopic. While it is not a perfect fit, we assigned it to the IPTC economy, business, and finance. While it is not about any of these things, there is a subtopic about housing.

Topic 6: Art Events**Top 5 words:** students, art, community, student, work**Top 2 stories:**

1. <https://johnjaysentinel.com/14320/campus-news/>
2. <https://richlandstudentmedia.com/11955/showcase/community-cultivates-at-cottonwood-art-festival/>

IPTC: Arts, Culture, Entertainment, and Media**Metatopic:** Events**CIN or Community Cohesion:** Community Cohesion**Campus words:** Yes

The top words were centered around campus events and people, with an emphasis on arts and history. Story 1 is about the African Student Association hosting a fashion show to showcase African culture. Story 2 is about a community art festival. Story 2 is not about campus, but many of the top words are very college driven (faculty, students, campus, college, etc.). Because of this we are labeling art events with an emphasis on colleges. Since there is an emphasis on college and events, over being fully about art, we put it in the events metatopic, not the art one. We did but it in the Arts, Culture, Entertainment, and Media IPTC topic over the Lifestyle and Leisure topic, where events could go, because the two stories and some top words are about art.

Topic 7: Sports (baseball)**Top 5 words:** game, run, inning, runs, home**Top 2 stories:**

1. <https://gcsommit.com/16005/sports/grossmont-baseball-recap-the-last-10-games/>
2. https://www.thedaonline.com/sports/baseball/mountaineers-historic-season-comes-to-an-end-in-omaha-falling-to-unc-in-semifinals/article_689a54f0-7947-4ed6-9ecb-3f03f394ef8c.html

IPTC: Sport**Metatopic:** Sports**CIN or Community Cohesion:** Community Cohesion**Campus words:** Yes (class grades)

The top words are almost exclusively about baseball, with a few being about sports in general or different grade levels. Story 1 is a recap of the college's last 10 baseball games. Story 2 is about how the school's baseball had a great season but lost in the semifinals. These stories both align with the top words being about baseball.

Topic 8: Assistance Programs**Top 5 words:** health, food, students, people, program**Top 2 stories:**

1. <https://dailyegyptian.com/124494/showcase/federal-cuts-are-hitting-illinois-food-assistance-recipients-and-the-farmers-who-feed-them/>

2. <https://blogs.farmingdale.edu/sites/the-dale/2025/08/28/the-changing-face-of-snap-a-look-at-americas-food-assistance-program-from-2022-2025/>

IPTC: Society

Metatopic: Health and Welfare

CIN or Community Cohesion: CIN

Campus words: No (says student(s), but that could be non-college students)

The top words are about food and mental health programs/community support as well as the economy and gas prices. Story 1 is about federal changes to SNAP and how they are hurting Illinois residents and the farmers that feed them. Story 2 is about how SNAP has changed from 2022-2025. Since both of the stories are about SNAP, this topic seems to mostly be about food assistance programs. Because the other top words related to different struggles many Americans are facing, we will call this topic assistance programs in general. Both the metatopic and IPTC labels are related to welfare.

Topic 9: Social Media

Top 5 words: media, people, social, students, women

Top 2 stories:

1. <https://bonfire.digital.uic.edu/opinion/opinion-eye-catching-instagram-trends-and-how-they-lead-to-body-dysmorphia/>
2. <https://sunnewsdaily.com/stories/articles/2026/04/28/utah-tech-seniors-preparing-for-graduation/>

IPTC: Arts, Culture, Entertainment, and Media

Metatopic: STEM

CIN or Community Cohesion: Community Cohesion

Campus words: Yes

The top words are about social media/technology, people and identity, mental health, and studies/research. Story 1 is about Instagram and how it created body image issues. It also cites specific studies and mentions additional harms of social media. Story 2 is a video and was incorrectly scrapped; it got some of the website and then an error. It does say the word content a fair amount which is a top word. The top words and Story 1 are all related to social media and its harms, so we will be ignoring Story 2.

Topic 10: Performance Art and Media

Top 5 words: music, film, show, movie, love

Top 2 stories:

1. <https://thehofstrachronicle.com/114503/arts-entertainment/track-by-track-review-you-seem-pretty-sad-for-a-girl-so-in-love/>
2. <https://everettclipper.com/18127/entertainment/>

IPTC: Arts, Culture, Entertainment, and Media

Metatopic: Art

CIN or Community Cohesion: Community Cohesion

Campus words: No

The top words are about music, movies/shows and performing. Besides words relating to albums, there are words about films, concerts, musicals, and acting. Story 1 is a track-by-track review of Olivia Rodrigo's third album. It discusses the songs, their meaning, and fans' reactions. Story 2 is a review of Noah Kahan's most recent album. It discusses the broad themes of the album and some specific songs. The 2 stories are about albums, but the top words are not just about music or recorded media. We decided on the label performance art and media, because that encompasses live and recorded content and different types of art.

Topic 11: Development (and months)

Top 5 words: data, science, associate, water, center

Top 2 stories:

1. <https://campuschroniclenews.com/sam-hosts-a-random-podcast-with-jayden-and-ben/>
2. <https://campuschroniclenews.com/jayden-and-makayla-recap-2026-fashion-show/>

IPTC: Environment

Meta: Environment and Planning

CIN or Community Cohesion: CIN

Campus words: Yes

The top words are a little all over the place. Some are about data centers and development while others are months or about campus. The two stories are both podcasts from the same publication. The text is mostly from the website which includes archives listed by month and year and other article titles. We will be basing this topic on the non-month related words, but it may make some articles that are empty or discussing months be labeled as CIN about data centers. The non-month words are also not in full agreement but seem to center around the environment and building.

Topic 12: Elections

Top 5 words: state, election, bill, public, city

Top 2 stories:

1. <https://sdcitytimes.com/news/2026/05/22/primary-26-porter-introduction/>
2. <https://ulmhawkeyeonline.com/38534/news/what-you-need-to-understand-before-primary-party-elections-begin/>

IPTC: Politics and Government

Metatopic: Political Life

CIN or Community Cohesion: CIN

Campus words: No

The top words are all related to politics. Most are about voting and candidates while some are on issues. Story 1 is about the primary election for governor in California and how one of the candidates will not take corporate donations. Story 2 is also about the primary party election, but in Louisiana and what voters need to know.

Topic 13: Law enforcement/Crime**Top 5 words:** law, ice, police, immigration, people**Top 2 stories:**

1. <https://dailyegyptian.com/124420/uncategorized/>
2. <https://longislandadvocate.com/nassau-countys-cooperation-with-ice-violates-the-law-attorneys-say/>

IPTC: crime, law and justice**Metatopic:** Emergencies and Risks**CIN or Community Cohesion:** CIN**Campus words:** No

The top words are about law enforcement, ICE, courts, and the current administration. Story 1 is about an Illinois man who was convicted after making 3D printed guns. Story 2 is about Nassau County and an attorney claiming cooperation with ICE is against state law. This topic is about crime, law enforcement, and courts.

Topic 14: Schools**Top 5 words:** school, rpi, schools, year, people**Top 2 stories:**

1. <http://www.talon.news/features/multi-year-exercise-science-study-comes-to-an-end/>
2. <https://sm2media.com/38180/news/pay-raise-brings-relief-but-not-a-solution-for-teacher-shortage/>

IPTC: Education**Metatopics:** Education**CIN or Community Cohesion:** CIN**Campus words:** Yes

The top words are about schools (multiple levels), teachers, communities, and a few miscellaneous words. Story 1 is about a three-year study done by seniors in their capstone ending. Story 2 is about raising teacher pay to increase the number of them. While this helped with the shortage, it did not solve it. The topic seems to generally be about schools at multiple levels and things that happen at them.

Topic 15: Food**Top 5 words:** food, people, day, students, coffee**Top 2 stories:**

1. <https://eldonnews.org/style/2026/06/03/maxs-munchies-how-to-elevate-simple-foods-on-a-college-budget/>
2. <https://lvcampustimes.org/2026/05/fair-food-falls-short/>

IPTC: lifestyle and leisure**Metatopic:** Community Events**CIN or Community Cohesion:** Community Cohesion**Campus words:** Yes

The top words are about businesses, mostly local, and food. Story 1 is advice on how to eat well on a college budget. Story 2 is about fair food at the LA county fair and how the writers and their friends ranked everything. This category is about food, but also local businesses and restaurants. We used the IPTC topic Lifestyle and Leisure because going out to eat is a leisurely activity, and there are also subcategories about food. It does not perfectly match any of the metatopics, but we went with community events because it is partially about community businesses. Story 2 does not fit this metatopic very well, but it does not fit any others better.

Topic 16: Nature, traveling, websites

Top 5 words: cookies, site, usc, website, google

Top 2 stories:

1. <https://theprairienews.com/34358/showcase/why-is-the-sky-blue/>
2. <https://dailytrojan.com/2026/06/10/for-the-record-april-9-2026-2/>

IPTC: None

Metatopic: Other

CIN or Community Cohesion: Mixed

Campus words: No

There are many unrelated top words. Some are about websites, some are about journalism, and some are about nature. There are also words about community and traveling. Story 1 is about the science behind why the sky is blue. Story 2 discusses 5 articles that were removed from the website due to AI use. Looking at the actual scraped text, we can see only the donation plug was collected. Some of the top words do seem to be about journalism, content, and words directly from the donation plug. Many of the environmental words are about birds, not the sky, which makes it more unclear why these words are grouped together. We will be categorizing this as none/other since a clear label cannot be decided on.

Topic 17: U.S. Politics and Religion

Top 5 words: trump, god, church, people, war

Top 2 stories:

1. <https://vikingfusion.com/2026/04/23/president-trumps-online-feud-with-pope-leo-threatens-catholic-approval/>
2. <https://eldonnews.org/style/2026/06/03/maxs-munchies-how-to-elevate-simple-foods-on-a-college-budget/>

IPTC: Politics and Government

Metatopic: Political Life

CIN or Community Cohesion: CIN

Campus words: No

The top words are about U.S. politics, specifically Trump and the War with Iran, and religion (Christian and Catholic). Story 1 is about Trump's feud with Pope Leo and how it may lose him Catholic support. Story 2 is about Missouri allowing alcohol to be sold on Sundays. The second story related less to the top words than Story 1 but is still about politics and has a slight religious

context because that is why certain activities were historically banned on Sundays. This topic is U.S. politics, but with an emphasis on religion. There is an IPTC topic for both politics and religion, but we went with the politics one because that is the main theme of this topic.

Topic 18: Experiences

Top 5 words: time, people, life, year, work

Top 2 stories:

1. <https://observer.case.edu/the-rose-i-chose/>
2. <https://langstongazette.com/yes-no-balance-how-to-make-smart-commitments-your-langston-experience/>

IPTC: Society

Metatopic: People/Relationships

CIN or Community Cohesion: Community Cohesion

Campus words: Yes

The top words are about experiences and people as well as college words. Story 1 is about the publications director of digital media graduating and how much the paper meant to her. Story 2 is about knowing when to say yes or no to have the best college experience. Both of these are about making the best of a certain time in your life. Combined with the top words, we are labeling it experiences, but it is primarily college ones. For metatopics, we were between community events or people/relationships. Because the top words are about what was taken away, and not a specific event, we went with people/relationships.

Topic 19: Sports (college)

Top 5 words: team, season, game, year, coach

Top 2 stories:

1. <https://iowastatedaily.com/339694/sports/after-the-cyclone-2025-26-iowa-state-athletics-recap/>
2. <https://cuatower.com/2026/06/every-cardinal-student-athlete-who-received-2025-26-honors-and-awards/>

IPTC: Sport

Metatopic: Sports

CIN or Community Cohesion: Community Cohesion

Campus words: Yes

The top words are about sports in general; multiple are names. Story 1 is a recap of the 2025-2026 athletic season for Iowa State University. Story 2 is about all of the student athletes that receive an award at Catholic University for 2025-2026. In addition to some of the top words, like NCAA, this topic has an emphasis on college sports.

Topic 20: Campus Elections/College in General

Top 5 words: students, student, campus, university, year

Top 2 stories:

1. <https://poly.rpi.edu/news/2026/03/grace-meehan-candidate-profile>
2. <https://trail.pugetsound.edu/?p=19453>

IPTC: Education

Metatopic: Campus

CIN or Community Cohesion: Community Cohesion

Campus words: Yes

The top words are all related to college/university campuses, leaders, and administrators. Story 1, which we had to read the full_text column of, is the interview responses for an undergraduate presidential candidate. Story 2 is a Letter to the Editor about the college running an unfair election for the new Vice President of Institutional Equity and Diversity. Both of these stories are about college elections, but the top words are a little broader.

Topic 21: Conferences and Horse Racing

Top 5 words: staff, rica, community, people, college

Top 2 stories:

1. <https://www.theprospectordaily.com/2026/05/11/>
2. <https://theseahawk.org/38345/thelatest/opinion-the-kentucky-derby-a-tradition-rooted-in-abuse/>

IPTC: None

Metatopic: Other

CIN or Community Cohesion: Community Cohesion

Campus words: Yes

The top words are not very coherent. Some are about horses, some are about location/college, and some are about programs/policies. Story 1 is about an Alzheimer's conference in El Paso, TX. Story 2 is an opinion piece about how the Kentucky Derby is rooted in horse abuse. These stories do make sense with many of the top words, but do not have much of a connection. While a clear label cannot be decided, the words and stories are community cohesion, not CIN.

Topic 22: Events (specifically ones for fun)

Top 5 words: event, students, club, events, student

Top 2 stories:

1. <https://theprairienews.com/34437/features/galaxycon-hosts-hayden-christensen-in-oklahoma-city/>
2. <https://thefalcon.seapacmedia.com/19131/features/annual-springtime-dance/>

IPTC: Lifestyle and Leisure

Metatopic: Community Events

CIN or Community Cohesion: Community Cohesion

Campus words: Yes

The top words are about events/activities, primarily on campus. Story 1 is about a Star Wars convention in Oklahoma City. Story 2 is about the annual spring ball hosted by the university. Since story 1 is not about a campus event, but a community one, we are labeling the topic events

in general. This topic could fit into multiple IPTC codes because it is regarding clubs (lifestyle and leisure), celebrations (human interest), and communities (society). We decided it was lifestyle and leisure because the events do not have to be a celebration and community felt broader than the topic.

Appendix D: IPTC Media Codes Figure and Analysis

Figure 3: The Average Proportion of Each IPTC Code for News partnership and LNA Categories

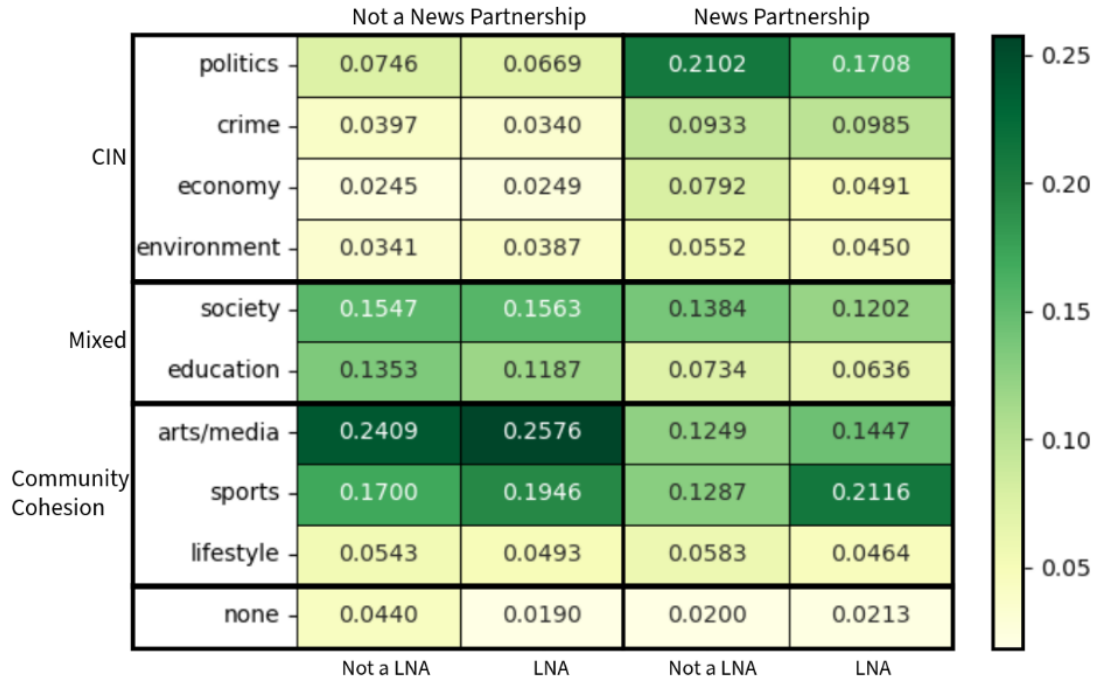


Figure 4 is a heatmap of the average topic proportions for each of the IPTC media codes we assigned topics to. The IPTC codes are divided into CIN, community cohesion, mixed, and none. The mixed category contains IPTC codes that were assigned to CIN topics and community cohesion topics. A higher average proportion is represented by dark green while a lower proportion is light yellow. There are four columns of proportions which from left to right are non-news partnership and non-LNA, non-news partnership and LNA, news partnership and non-LNA, and news partnership and LNA.

Figure 4 shows that non-news partnerships rarely discuss CIN and frequently discuss mixed and community cohesion topics. Regardless of if the articles were in or adjacent to a LNA, non-news partnerships primarily discuss Society, Education, Arts/media, and Sports. As with Figure 3, there is very little difference between LNA and non-LNA articles that are not a part of a news partnership.

News partnerships do discuss CIN more than non-news partnerships, but this difference is not as extreme as with our metatopics. The only CIN topic that is discussed at a high proportion is Politics, but Society is also frequently discussed and contains CIN topics.

Arts/media and Sports are also frequently discussed. Non-LNA discuss Politics more than LNA do while LNA discuss Sports more. The IPTC codes offer additional information regarding our analysis but are less specific than our metatopics.

Supplementary Material:

The original dataset is from UVM's Center for Community News (CCN) and is a private dataset of the CCN's Hugging Face. <https://huggingface.co/center-for-community-news>

The datasets and code created for this research project can be found at

<https://github.com/BellaBOlinger/SOCKS-2026-News>.

All additional datasets can be found on the CCN's GitHub Repository. <https://github.com/UVM-CCN/ccn-data>